ACCENTURE INNOVATION CHALLENGE 2026 • OFFICIAL TECHNICAL README

ControlPlane.ai

Model-Agnostic Real-Time AI Oversight, Guardrail & Quality Control Tower

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1. Executive Overview & Problem Statement

As organizations rush Generative AI into customer support, coding, executive decision-making, and internal workflows, one critical question stops being theoretical: **What happens the moment an AI gives an answer that sounds right — but isn't?**

AI systems can be confidently wrong, generate misleading answers with authoritative tone, quietly burn compute through unnecessary tokens, and expose organizations to sensitive-data leakage without a single traditional infrastructure alarm going off.

$67.4 Billion

HALLUCINATION BUSINESS LOSS

Estimated global enterprise cost of AI hallucinations in 2024 (AllAboutAI / Korra Report).

97%

SECURITY ACCESS FAILURES

Of organizations reporting AI security breaches lacked real-time inline access guardrails (IBM 2025 Study).

63%

LACK FORMAL GOVERNANCE

Enterprises deploy LLMs to production without automated quality gates, relying solely on uptime logs (Deloitte).

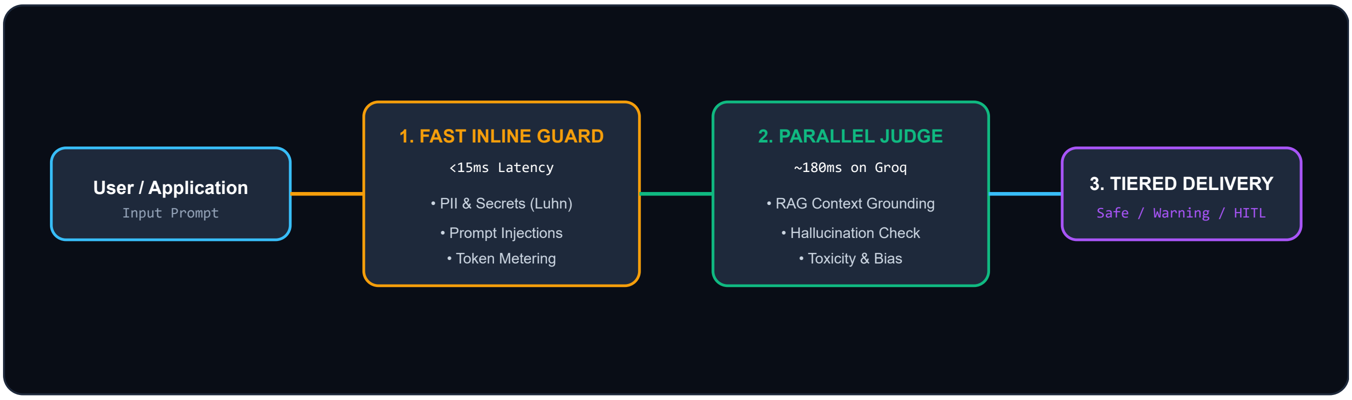
< 15 ms

FAST INLINE SLA

Synchronous PII and injection inspection intercepts attacks without adding user lag.

2. System Architecture: The Dual-Speed Engine

ControlPlane.ai recognizes that **not all guardrails need to run at the same speed**. Fast signals run synchronously inline, while deep judge models verify factual groundedness in parallel.



3. Why We Used These Technologies (Design Rationale)

TECHNOLOGY	LAYER	ENGINEERING RATIONALE & STRATEGIC WHY
Groq Cloud LPUs	Inference Engine	Provides <200ms time-to-first-token inference for 70B and 8B models. Enables live LLM-as-a-Judge evaluation in parallel with zero perceivable delay to the end user.
Meta Llama 3 & Compound Models	Base & Judge LLM	Open-weights, enterprise-safe, with superior reasoning on structured JSON output and factual comparison benchmarks.
FastAPI (Python)	Backend Gateway	Asynchronous native ASGI framework with built-in Pydantic validation, concurrent background task workers, and high-performance WebSockets.
WebSockets (Native)	Live Telemetry	Instant bidirectional push of telemetry logs to the Control Tower UI with sub-millisecond overhead compared to polling.
React 19 + TypeScript + Vite	Frontend Architecture	Strict type safety across the entire telemetry contract, rapid HMR developer experience, and component-level re-rendering optimization.
Tailwind CSS (Dark Obsidian)	UI / UX Design	High-contrast, accessibility-compliant executive dark aesthetic (`#07090E`) with glowing status indicators.
Recharts	Data Visualization	Declarative, GPU-accelerated SVG charting for OWASP threat distribution and latency histograms.

4. OWASP Top 10 for LLMs Security Mapping

OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
`LLM01`	**Prompt Injection**	Synchronous heuristic and delimiter analysis stops jailbreak attempts in <15ms.
`LLM02`	**Sensitive Info Disclosure**	Regex, Luhn verification, and secret scanners strip/block API keys, credit cards, and SSNs.
`LLM03`	**Hallucination / Misinformation**	Parallel Judge model scores groundedness against verified reference documents.
`LLM04`	**Data & Model Poisoning**	HITL audit trail isolates anomalous model outputs and triggers administrative review.
`LLM06`	**Excessive Agency**	Dynamic Tiered gating prevents unauthorized downstream execution of unverified commands.

OWASP ID	VULNERABILITY NAME	CONTROLPLANE.AI DEFENSIVE MECHANISM
`LLM10`	**Unbounded Consumption**	Token metering and adaptive sampling depth caps prevent denial-of-wallet compute spikes.

5. 90-Second Pitch & Live Demo Script

1. **The Problem (0:00 – 0:30 on Home / Control Tower)**: Explain that traditional APMs only watch uptime and latency — they cannot detect when an AI is confidently wrong. In 2024, hallucinations cost enterprises $67.4B.
2. **The Dual-Speed Engine (0:30 – 1:05 on Gateway Lab)**:
   - Click **Preset 1 (Credit Card Exfiltration)**: Show Fast Guard intercepting sensitive PII in 1.2ms.
   - Click **Preset 2 (Financial Hallucination)**: Show Parallel Judge catching the fabricated 34.8% growth metric and attaching a Citation Warning.
3. **Threat Surge & HITL Audit (1:05 – 1:30 on Control Tower & HITL Queue)**:
   - Click **Threat Surge**: Watch the Adaptive Sampling gauge auto-climb from 25% to 85% to shield infrastructure.
   - Open **HITL Queue**: Demonstrate 1-click remediation and export of timestamped compliance packages.